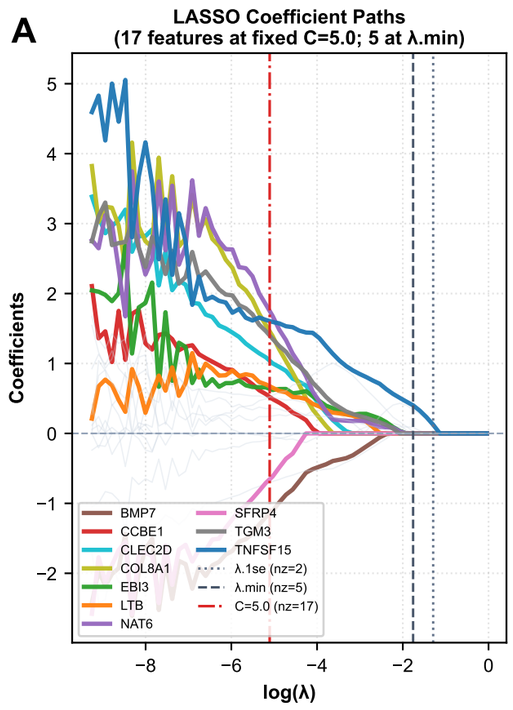
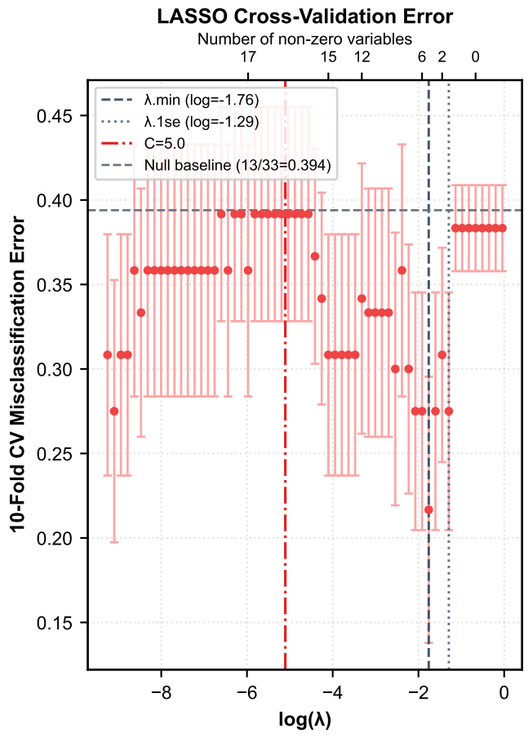
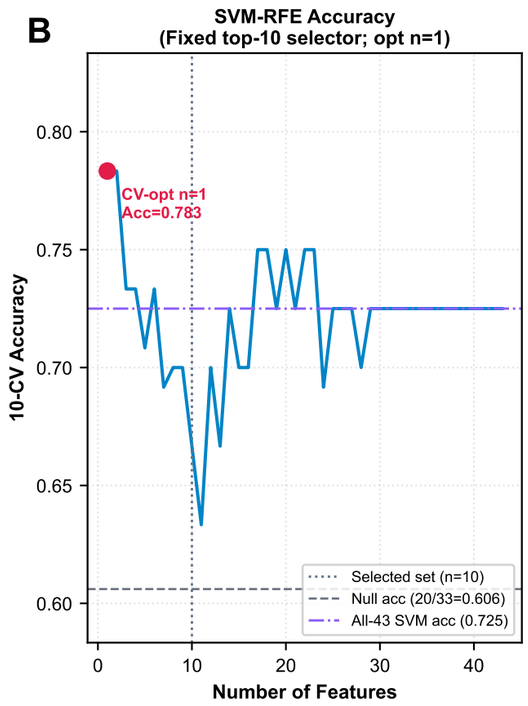
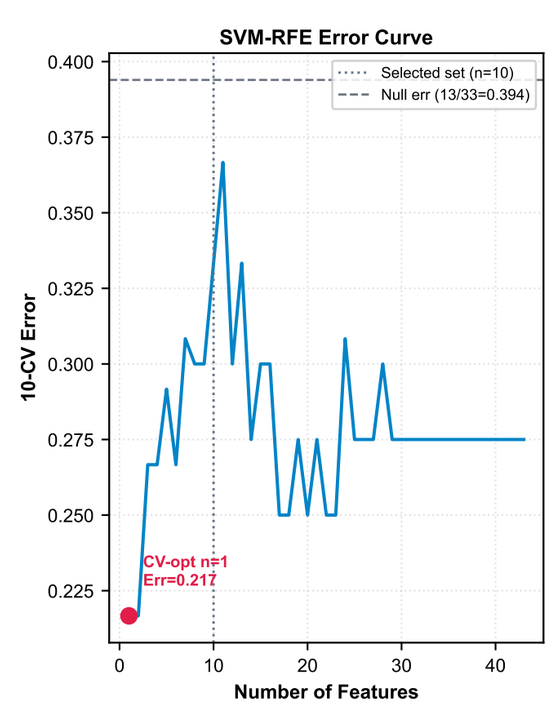
A


LASSO Coefficient Paths
(17 features at fixed C=5.0; 5 at λ.min)
Coefficients
log(λ)
BMP7
CCBE1
CLEC2D
COL8A1
EBI3
LTB
NAT6
SFRP4
TGM3
TNFSF15
λ.1se (nz=2)
λ.min (nz=5)
C=5.0 (nz=17)

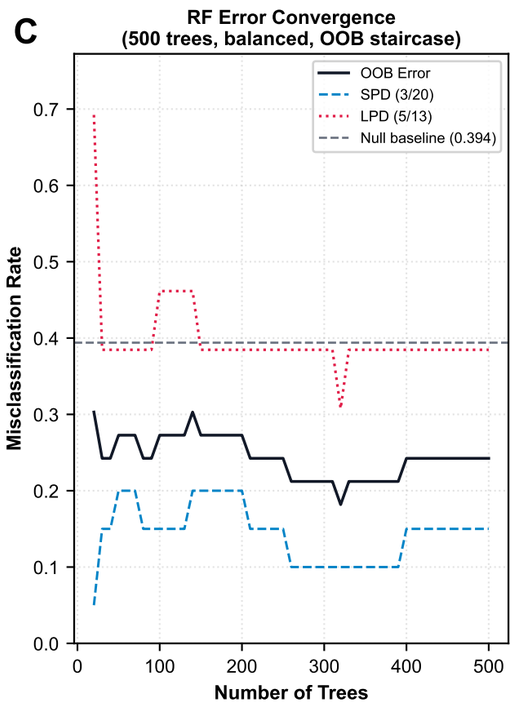

LASSO Cross-Validation Error
Number of non-zero variables
17 15 12 6 2 0
λ.min (log=-1.76)
λ.1se (log=-1.29)
C=5.0
Null baseline (13/33=0.394)
10-Fold CV Misclassification Error
log(λ)

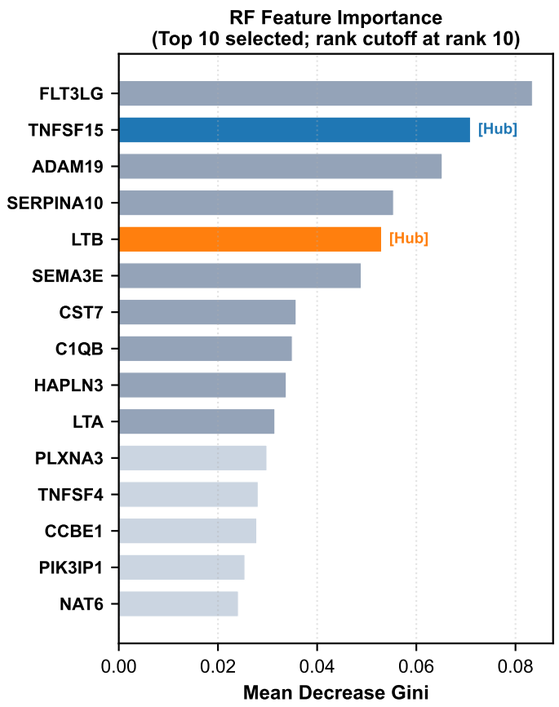
B


SVM-RFE Accuracy
(Fixed top-10 selector; opt n=1)
CV-opt n=1
Acc=0.783
10-CV Accuracy
Number of Features
Selected set (n=10)
Null acc (20/33=0.606)
All-43 SVM acc (0.725)

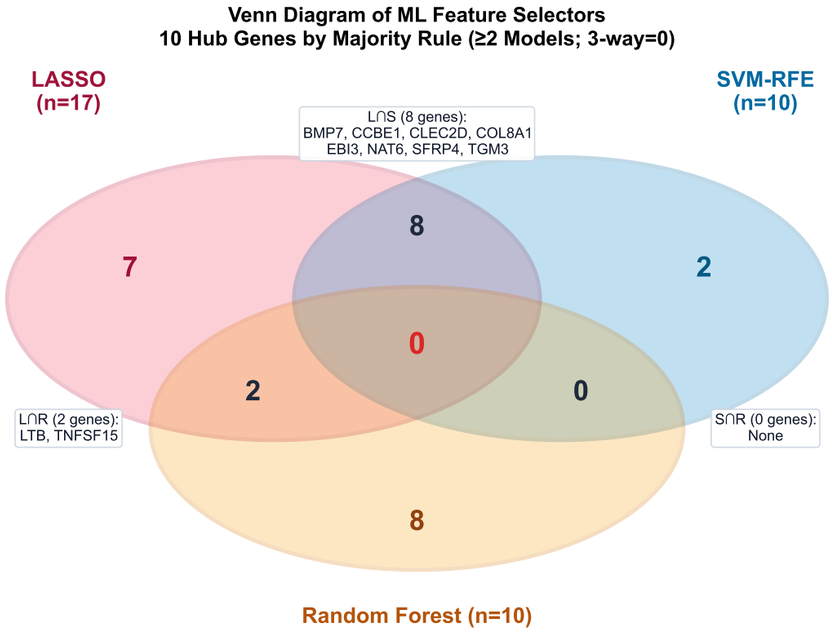

SVM-RFE Error Curve
Selected set (n=10)
Null err (13/33=0.394)
CV-opt n=1
Err=0.217
10-CV Error
Number of Features

C

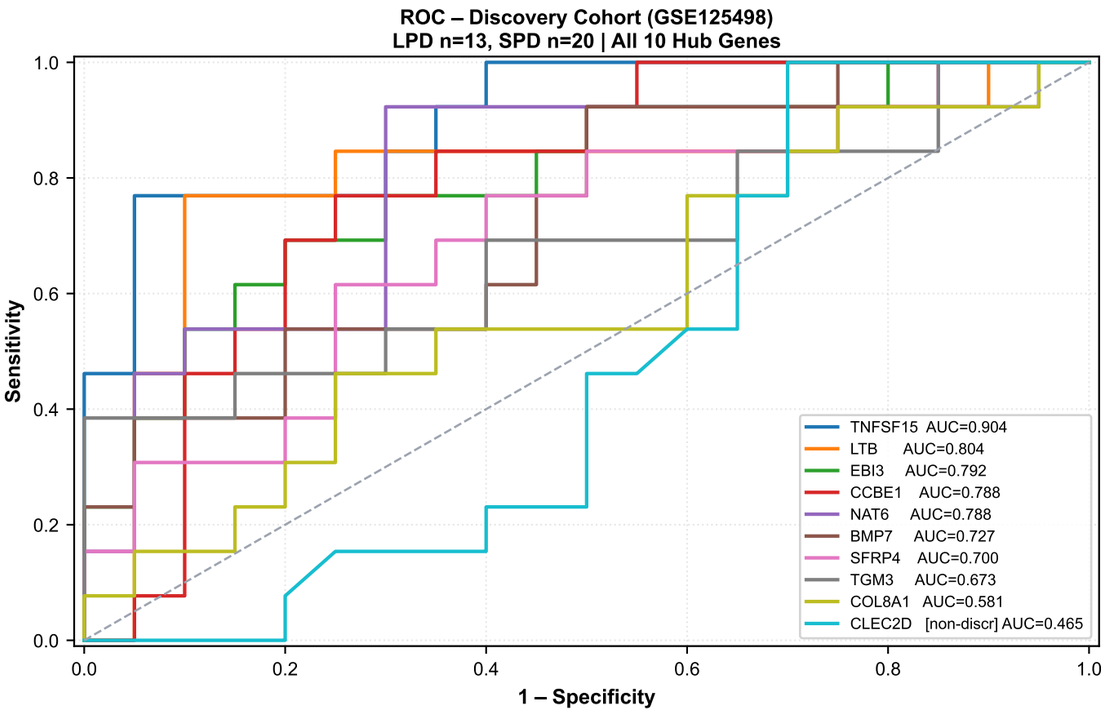

RF Error Convergence
(500 trees, balanced, OOB staircase)
OOB Error
SPD (3/20)
LPD (5/13)
Null baseline (0.394)
Misclassification Rate
Number of Trees


RF Feature Importance
(Top 10 selected; rank cutoff at rank 10)
FLT3LG
TNFSF15 [Hub]
ADAM19
SERPINA10
LTB [Hub]
SEMA3E
CST7
C1QB
HAPLN3
LTA
PLXNA3
TNFSF4
CCBE1
PIK3IP1
NAT6
Mean Decrease Gini

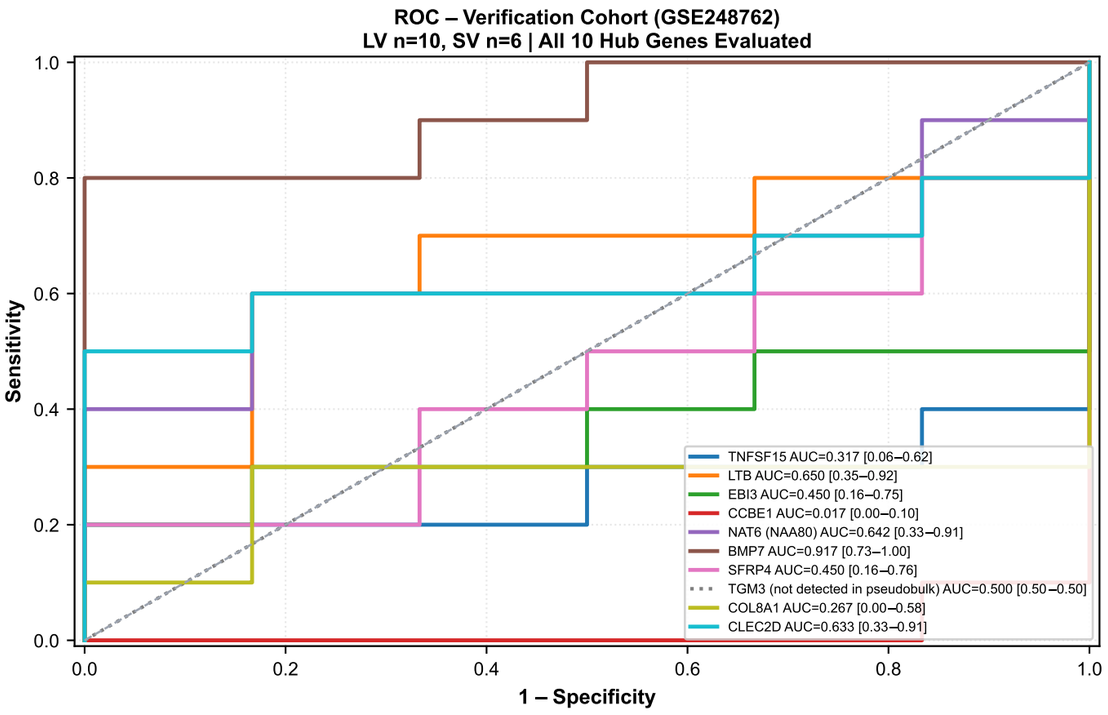
D


Venn Diagram of ML Feature Selectors
10 Hub Genes by Majority Rule (≥2 Models; 3-way=0)
LASSO
(n=17)
SVM-RFE
(n=10)
LΩS (8 genes):
BMP7, CCBE1, CLEC2D, COL8A1
EBI3, NAT6, SFRP4, TGM3
7
8
2
0
2
0
LΩR (2 genes):
LTB, TNFSF15
SΩR (0 genes):
None
8
Random Forest (n=10)

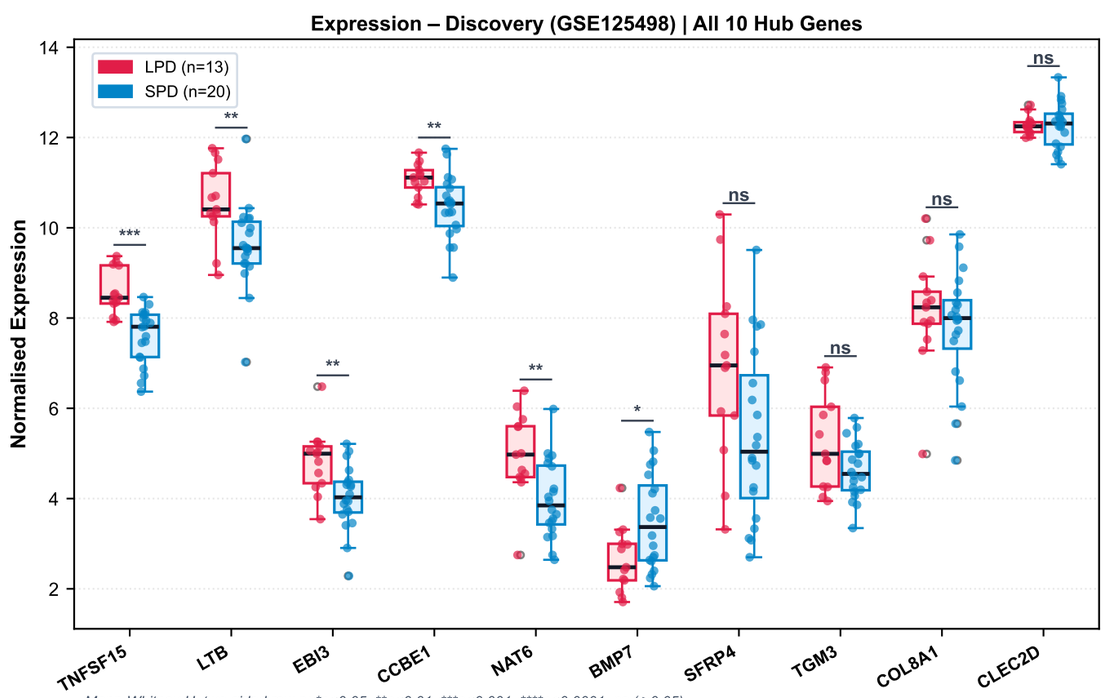
E


ROC – Discovery Cohort (GSE125498)
LPD n=13, SPD n=20 | All 10 Hub Genes
Sensitivity
1 – Specificity
TNFSF15 AUC=0.904
LTB AUC=0.804
EBI3 AUC=0.792
CCBE1 AUC=0.788
NAT6 AUC=0.788
BMP7 AUC=0.727
SFRP4 AUC=0.700
TGM3 AUC=0.673
COL8A1 AUC=0.581
CLEC2D [non-discr] AUC=0.465

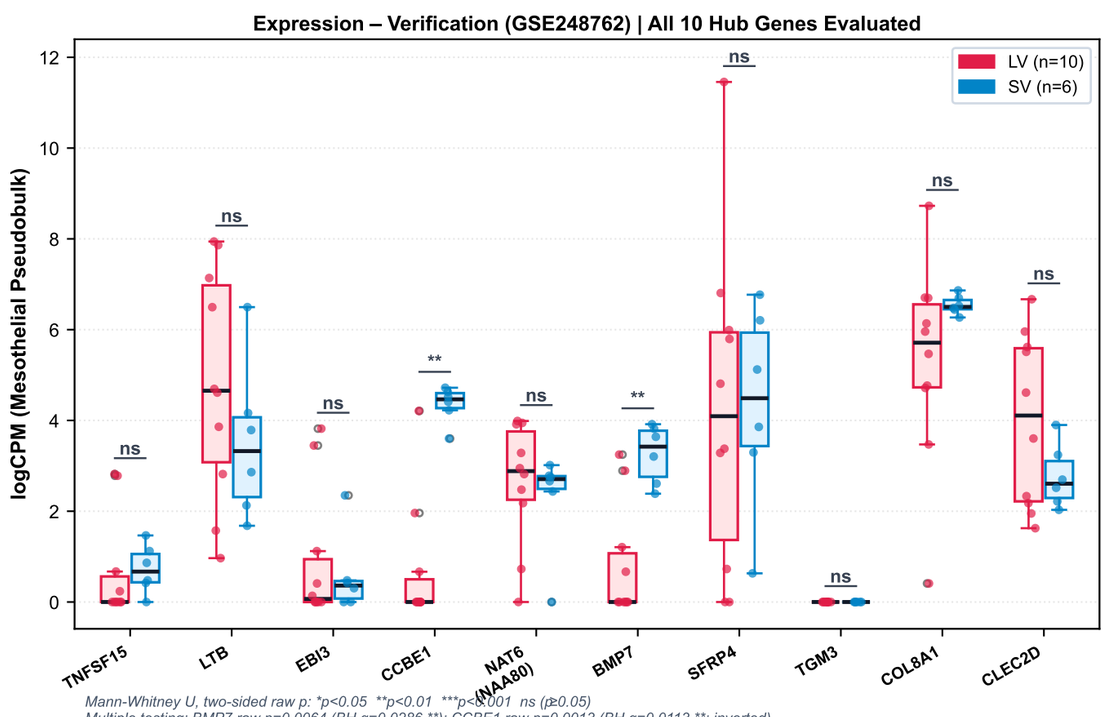
F


ROC – Verification Cohort (GSE248762)
LV n=10, SV n=6 | All 10 Hub Genes Evaluated
Sensitivity
1 – Specificity
TNFSF15 AUC=0.317 [0.06–0.62]
LTB AUC=0.650 [0.35–0.92]
EBI3 AUC=0.450 [0.16–0.75]
CCBE1 AUC=0.017 [0.00–0.10]
NAT6 (NAA80) AUC=0.642 [0.33–0.91]
BMP7 AUC=0.917 [0.73–1.00]
SFRP4 AUC=0.450 [0.16–0.76]
TGM3 (not detected in pseudobulk) AUC=0.500 [0.50–0.50]
COL8A1 AUC=0.267 [0.00–0.58]
CLEC2D AUC=0.633 [0.33–0.91]

G


Expression – Discovery (GSE125498) | All 10 Hub Genes
LPD (n=13)
SPD (n=20)
Normalised Expression
TNFSF15
LTB
EBI3
CCBE1
NAT6
BMP7
SFRP4
TGM3
COL8A1
CLEC2D
Mann-Whitney U, two-sided raw p: *p<0.05 **p<0.01 ***p<0.001 ****p<0.0001 ns (p≥0.05)
Note: BMP7 raw p=0.0311 (*) but Benjamini-Hochberg q=0.0519 (not significant after FDR)

H


Expression – Verification (GSE248762) | All 10 Hub Genes Evaluated
LV (n=10)
SV (n=6)
logCPM (Mesothelial Pseudobulk)
TNFSF15
LTB
EBI3
CCBE1
NAT6 (NAA80)
BMP7
SFRP4
TGM3
COL8A1
CLEC2D
Mann-Whitney U, two-sided raw p: *p<0.05 **p<0.01 ***p<0.001 ns (p≥0.05)
Multiple testing: BMP7 raw p=0.0064 (BH q=0.0286 **); CCBE1 raw p=0.0013 (BH q=0.0113 **; inverted)